Field descent and compatible tori

Human Compression project

Lang–Steinberg descent gives a common representative for pairwise incident cosets of subgroups fixed by independent field automorphisms. Cyclic coset coordinates then add a torus normalizer as another type. We prove the resulting Suzuki geometries are thick, residually connected and residually weakly primitive in every rank. The proof retains classical Suzuki structure and Lang–Steinberg, and includes the exact finite matrices used to choose compatible tori.

1 Main statements

The previously acquired rank-four configuration, described by the explicit groups below, is a flag-transitive geometry. It is firm and residually connected as well as RWPRI. The argument below establishes the common-point property for every flag, without enumerating the ambient $\mathrm{Sz}(2^{105})$.

There is also an unconditional family in every rank. Define, for positive odd d,

$$\tau_d = 2^d + 1 - \big((1+i)^d + (1-i)^d\big). \tag{1}$$

Thus $\tau_1 = 1, \tau_3 = 13, \tau_5 = 41, \tau_7 = 113, \tau_9 = 481$. Equivalently, $\tau_d = 2^d + 1 - \epsilon_d 2^{(d+1)/2}$, where $\epsilon_d = 1$ for $d \equiv 1, 7 \pmod 8$, and $\epsilon_d = -1$ for $d \equiv 3, 5 \pmod 8$.

Theorem A. Fix odd $e \geq 3$, and any distinct odd primes $a_1, \ldots, a_k$ not dividing τ_e. For each i, choose any prime divisor

$$\ell_i \mid \tau_{ea_i}/\tau_e.$$

These quotients are integers greater than one. Their prime divisors are coprime to τ_e and distinct between different indices. Put $n = e\prod_i a_i$, $m = \tau_e \prod_i \ell_i$, and $G = \mathrm{Sz}(2^n)$. There are simultaneous standard subfield subgroups

$$S_i = \mathrm{Sz}(2^{n/a_i})$$

and one subgroup $K = C_m : C_4$ for which $\Gamma(G; S_1, \ldots, S_k, K)$ is a thick, residually connected, flag-transitive RWPRI geometry of rank $k+1$. The theorem provides explicit intersection formulas in Section 6.

In particular, fixing $e = 3$ and choosing any k distinct odd primes other than 13 gives a geometry of rank $k+1$. No conjecture about infinitely many prime values of Suzuki torus orders is used.

For example, with $e = 3$ and relative field primes $3, 5, 7, 11$, one can choose $\ell_i = 37, 41, 113, 2113$. This gives a rank-five example over $\mathbb{F}_{2^{3465}}$, described by the same formulas as the smaller ones. Enumerating this field or group is unnecessary.

The construction specializes the nonsplit-torus pattern in Construction 9 of L110, Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, pp.17–18. It supplies simultaneous embeddings and flag transitivity for this family. It is not a replacement classification of all the paper's configurations or all relative positions of its subgroup types.

Two reusable principles do most of the geometric work:

- compatible cosets of fixed subgroups of coprime field automorphisms have a common point, by Lang descent;
- a subgroup with independent coset coordinates can be added as another type without losing that common-point property.

2 Cyclic descent and intersections of subfield cosets

Let $\mathbf{H}$ be a connected algebraic group over an algebraic closure of a finite field, and let F be a Steinberg endomorphism. We use Lang–Steinberg surjectivity: every $u \in \mathbf{H}$ has the form $x^{-1}F(x)$. This is retained classical theory, not reproved here; see, for example, Harper, *Totally deranged elements of almost simple groups and invariable generating sets*, Section 2.2 and equation (2.1), which use it for Steinberg endomorphisms, including the very twisted groups.

Lemma B: finite cyclic descent. Set $G = \mathbf{H}^{F^m}$, and let $\sigma = F^d|_G$, where $d \mid m$ and $h = m/d$. If $u \in G$ satisfies

$$u\,\sigma(u)\cdots\sigma^{h-1}(u) = 1,$$

then $u = x^{-1}\sigma(x)$ for some $x \in G$.

Indeed, apply Lang–Steinberg to F^d on the connected algebraic group: choose $x \in \mathbf{H}$ with $u = x^{-1}F^d(x)$. The displayed norm telescopes to $x^{-1}F^m(x)$. Its being one proves $x \in G$, which is precisely the finite-field conclusion needed. No coprimality with the characteristic is required.

Equivalently, in $G \rtimes C_h$, with the specified cyclic action, every element projecting to a generator and having order h is G-conjugate to that generator. The cyclic factor here may be treated as an abstract group of order h, so no faithfulness issue is hidden.

Theorem C: intersections of subfield cosets. Suppose $b_1, \ldots, b_k$ are pairwise coprime divisors of m. Define

$$S_i = \mathbf{H}^{F^{m/b_i}} \le G = \mathbf{H}^{F^m}.$$

For any selection of indices, if the left cosets g_iS_i intersect pairwise, then their total intersection is nonempty.

Proof. Work in $E = G \rtimes \langle f \rangle$, with f an abstract element of order m acting as $F|_G$, and put $f_i = f^{m/b_i}$. Associate to g_iS_i the element

$$c_i = g_i f_i g_i^{-1}.$$

This association is well defined: $S_i = C_G(f_i)$. If g_iS_i and g_jS_j meet at y, both c_i, c_j are conjugated from f_i, f_j by y. Hence they commute. Their orders b_i are pairwise coprime, so the group D that they generate is cyclic of order $b = \prod_i b_i$. Its projection to $\langle f \rangle$ is an isomorphism onto $\langle f^{m/b} \rangle$.

Let c be its unique element projecting to $f^{m/b}$. Write $c = uf^{m/b}$. Since $c^b = 1$, its coefficient u satisfies the norm condition in Lemma B, with $d = m/b$. Consequently D is conjugate by an element $x \in G$ to $\langle f^{m/b} \rangle$. The conjugating element may be oriented so that

$$c_i = x f_i x^{-1} \quad \text{for every } i.$$

There is no ambiguity of generators: conjugation by G acts trivially on the quotient E/G, and the projection is injective on D. Thus $g_i^{-1}x$ centralizes f_i, so $x \in g_iS_i$ for every i. This proves the theorem.

For Suzuki groups take $\mathbf{H} = \mathrm{Sp}_4(\overline{\mathbb{F}}_2)$ and its standard exceptional Steinberg endomorphism ρ, with $\rho^2 = \phi_2$, ordinary binary Frobenius. For positive odd d, $\mathbf{H}^{\rho^d} = \mathrm{Sz}(2^d)$ in compatible standard coordinates. This is the classical Suzuki construction; see De Medts–Naert, *Suzuki–Ree groups as algebraic groups over* $\mathbb{F}_{\sqrt{p}}$, Section 2, for the exceptional endomorphism and its square. On $\mathbf{H}^{\rho^n}$, n odd, ρ and ordinary binary Frobenius generate the same cyclic field-automorphism group. Taking $F = \rho^e$, $m = \prod_i a_i$, therefore gives Theorem C for the standard S_i in Theorem A.

3 Adding a type by independent coset coordinates

The following lemma is purely group theoretic. In particular it does not identify a coset set with a quotient group.

Lemma D. Let (S_i) be a typed coset family in a group G with the common-point property of Theorem C. Suppose $K \leq G$, and put $H_i = K \cap S_i$. If the set map

$$K \longrightarrow \prod_i K/H_i, \qquad k \longmapsto (kH_i)_i \tag{2}$$

is surjective, then the enlarged typed coset family (S_i, K) also has the common-point property.

A flag not containing a K-coset is handled by the original property. Otherwise translate its K-coset to K. Each other coset meets K, so it is $k_i S_i$ with $k_i \in K$, and its intersection with K is $k_i H_i$. Surjectivity in (2) supplies a common element. Partial selections of types cause no problem: assign arbitrary cosets to the unused coordinates. Every flag therefore has a common representative and extends to a full standard-chamber translate. This proves the geometry axiom and flag transitivity.

A convenient sufficient condition is

$$K = (V_1 \times \cdots \times V_k) \rtimes B, \qquad H_i = \Big(\prod_{j \neq i} V_j\Big) \rtimes B, \tag{3}$$

where B preserves each direct factor. The restricted set map from $\prod_i V_i$ to the product of coset sets is a bijection: its i-th coordinate is just the V_i-coordinate. Neither H_i nor B needs to be normal in K.

This already settles the earlier rank-four example. There $V = C_{13} \times C_{41} \times C_{113}$, $B = C_4$, and the exact certificate gives precisely (3). The three standard subfield coset types satisfy Theorem C. Hence the four-type coset system is a flag-transitive geometry. Its RWPRI proof was already given there. Firmness and residual connectedness also follow from the argument in Section 7 below, with base $\mathrm{Sz}(2)$ and the three full prime-order tori 13, 41, 113.

4 Compatible tori in every odd degree

Here is the uniform construction behind the field-compatible cyclic groups, rather than an extrapolation from small computations.

Use the binary matrices M, r displayed in the immutable earlier note, where $M^4 + M + I = 0$, $r = U(1,0)$, $|r| = 4$, and $r^{-1}Mr = M^2$. The polynomial $X^4 + X + 1$ is irreducible. Thus $\mathbb{F}_2[M]$ is the field with sixteen elements.

For reference, the acquired seed matrices are

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}.$$

The order-thirteen seed, in $\mathbb{F}_8 = \mathbb{F}_2[z]/(z^3+z+1)$, is

$$a = \begin{pmatrix} z^2+z+1 & 0 & 1 & z^2+1 \\ 1 & z & z^2 & 1 \\ z^2 & z^2 & z+1 & z^2+1 \\ z^2 & 1 & z^2 & z+1 \end{pmatrix}.$$

These are the literal previously checked matrices. The retained checks verify group membership, order, centralization, the exceptional endomorphism identities, and both twenty-element binary centralizers; no search is needed to verify these displayed identities.

The exact order-13 matrix $a \in \mathrm{Sz}(8)$ found in the earlier seed commutes with M. Since the multiplicative order of 8 modulo 13 is four, every nontrivial order-13 matrix in dimension four over $\mathbb{F}_8$ has irreducible minimal polynomial of degree four. In particular a has four distinct eigenlines. The same is true of M: its degree-four binary polynomial remains irreducible over $\mathbb{F}_8$, since $\gcd(4,3) = 1$. Both therefore generate the same degree-four algebra over $\mathbb{F}_8$, and their commuting regular matrices have the same general linear centralizer. Put

$$\mathbf{T} = C_{\mathbf{H}}(M) = C_{\mathbf{H}}(a).$$

As the centralizer in Sp_4 of this regular semisimple element, $\mathbf{T}$ is a connected maximal torus of dimension two. This can also be seen directly in the eigenbasis of a: its symplectic eigenvalue pairs are inverse pairs, and the centralizer consists of the corresponding two independent diagonal coordinates.

Because M is binary, ϕ_2 preserves $\mathbf{T}$. Because $\rho^3(a) = a$, one has $\rho(a) = \rho^4(a) = \phi_2^2(a)$. This is another regular element of $\mathbf{T}$, so ρ also preserves $\mathbf{T}$.

Let B be the integral endomorphism induced by ρ on the rank-two character lattice of $\mathbf{T}$. Binary Frobenius permutes the four eigenlines of M in a four-cycle, since its minimal polynomial has degree four over $\mathbb{F}_2$. The corresponding action on the four symplectic weights $\{\chi_1, \chi_2, -\chi_1, -\chi_2\}$ is a rotation w of order four, with trace zero and determinant one. Therefore

$$B^2 = 2w, \qquad (\operatorname{tr} B)^2 - 2\det B = 0, \qquad (\det B)^2 = 4.$$

Integrality forces $\det B = 2$ and $\operatorname{tr} B = \pm 2$. The remaining sign is fixed by a tiny exact binary check:

$$\mathbf{T}^{\rho} = C_{\mathrm{Sz}(2)}(M) = 1.$$

The retained seed checker verifies this against all twenty matrices of $\mathrm{Sz}(2) = \langle r, w \rangle$. Equivalently, its unique order-five subgroup is generated by $r^2 w$, which does not commute with M.

For a torus, the fixed-point count for an endomorphism whose character matrix is B is $|\det(I - B)|$ when this determinant is odd. This particular count needs only Smith normal form: diagonalizing $I - B$ over the integers turns its kernel on $(\overline{\mathbb{F}}_2^{\times})^2$ into a product of groups of roots of unity of odd orders. Here the two sign choices give one or five fixed points, so the check selects $\operatorname{tr} B = 2$.

The eigenvalues of B are consequently $1+i, 1-i$. The same Smith-normal-form argument gives, for every positive odd d,

$$|\mathbf{T}^{\rho^d}| = \det(I - B^d) = 2^d + 1 - \left((1+i)^d + (1-i)^d\right) = \tau_d.$$

These are the nonsplit Suzuki torus orders. We retain the classical fact that such finite tori are cyclic and their normalizers in the simple Suzuki groups have order $4\tau_d$ and are maximal. Write $T_d = \mathbf{T}^{\rho^d}$. The binary r normalizes $\mathbf{T}$, belongs to every $\mathrm{Sz}(2^d)$, and has order four.

Thus $\langle T_d, r\rangle = T_d : C_4$ is that full normalizer for $d > 1$. For $d = 1$, $T_1 = 1$ and $\langle r\rangle = C_4$ has index five in $\mathrm{Sz}(2)$; the simple-group normalizer assertion is not being applied in this exceptional case.

The benefit is simultaneous position. For all positive odd a, b,

$$T_a \cap T_b = T_{\gcd(a,b)}, \qquad \gcd(\tau_a, \tau_b) = \tau_{\gcd(a,b)}. \tag{4}$$

The first equality follows from the fixed-point equations and Euclid's algorithm. For the second, place both cyclic groups in $T_{\mathrm{lcm}(a,b)}$, then use uniqueness of subgroups of each order in a cyclic group.

4.1 Exact matrix convention for the exceptional endomorphism

There is an elementary way to check that the exceptional endomorphism used above matches the earlier matrices. Take the symplectic form with antidiagonal matrix J. On the exterior square of the natural module, quotient the kernel of symplectic contraction by its invariant line $\langle e_1 \wedge e_4 + e_2 \wedge e_3\rangle$. The exterior-product pairing descends to a nondegenerate alternating form on this quotient, with antidiagonal matrix in the basis below. In the quotient basis

$$e_1 \wedge e_2,\ e_1 \wedge e_3,\ e_2 \wedge e_4,\ e_3 \wedge e_4,$$

the induced matrix $\rho(g)$ consists of the 2×2 minors whose row and column pairs are $12, 13, 24, 34$. This is the exceptional isogeny in the given coordinates. For four independent variables, let

$$u(a, A, b, B) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ aA + b & A & 1 & 0 \\ a^2A + ab + B & b & a & 1 \end{pmatrix}.$$

Direct polynomial multiplication gives

$$\rho(u(a, A, b, B)) = u(A, a^2, B, b^2). \tag{7}$$

Consequently ρ^2 is entrywise squaring on this full unipotent group. The diagonal matrix $\mathrm{diag}(x, y, y^{-1}, x^{-1})$ maps to $\mathrm{diag}(xy, x/y, y/x, (xy)^{-1})$, giving the same square; the antidiagonal w is fixed. These groups generate the symplectic group, so the identities recover $\rho^2 = \phi_2$. The exterior-square construction makes ρ a group homomorphism; the polynomial checks are not a test of that property on a sample.

Setting $A = a^\theta, B = b^\theta$, with $\theta^2 = 2$ on the finite field, equation (7) becomes exactly $\rho(U(a, b)) = U(a, b)^\theta$, with entrywise exponent on the right. The earlier diagonal and antidiagonal generators satisfy the same fixed-point relation. This binds the usual exceptional Suzuki construction to those particular matrices. The retained checker proves (7) as an identity of sparse polynomials over $\mathbb{F}_2$, checks the symplectic identity in four independent variables, and checks the diagonal identities as Laurent-polynomial identities. It also directly verifies $\rho(a) = a^{(4)}$, $\rho^3(a) = a$, and all twenty binary centralizer tests for the actual seed.

5 The prime choices exist without a prime-value conjecture

Let $\tau'_d = 2^d + 1 + ((1 + i)^d + (1 - i)^d)$. For odd d,

$$\tau_d \tau'_d = 2^{2d} + 1, \qquad \gcd(\tau_d, \tau'_d) = 1. \tag{5}$$

Indeed the trace square is 2^{d+1}, while the difference between the two odd factors is a power of two.

Fix odd $e \geq 3$ and an odd prime $a \nmid \tau_e$. Since $T_e \leq T_{ea}$, $\tau_e \mid \tau_{ea}$. The quotient is greater than one from (1), for example by bounding τ_{ea} below by $2^{ea} + 1 - 2^{(ea+1)/2}$ and τ_e above by $2^e + 1 + 2^{(e+1)/2}$.

If an odd prime $\ell \mid \tau_e$, the elementary lifting-the-exponent identity applied to $(2^{2e})^a + 1$ gives

$$v_\ell(2^{2ea} + 1) = v_\ell(2^{2e} + 1) + v_\ell(a) = v_\ell(\tau_e).$$

The last equality uses $a \nmid \tau_e$ and (5). Divisibility $\tau_e \mid \tau_{ea}$ forces exactly that same valuation in τ_{ea}. Thus

$$\gcd(\tau_{ea}/\tau_e, \tau_e) = 1.$$

Every prime divisor of the nontrivial quotient is therefore new. For distinct a_i, a_j, equation (4) gives $\gcd(\tau_{ea_i}, \tau_{ea_j}) = \tau_e$, so selected new primes cannot repeat. This proves all the prime-choice assertions of Theorem A. There is no appeal to Zsigmondy's theorem and no exceptional case hidden in an assumed primitive-prime-divisor statement.

6 Exact subgroup intersections and RWPRI

Let $P = \prod_i a_i$, $n = eP$, and choose the cyclic subgroup $A \leq T_n$ of order $m = \tau_e \prod_i \ell_i$. It exists because each factor divides τ_n and the factors are pairwise coprime. Put $K = A : \langle r \rangle$. The action of r preserves every subgroup of the cyclic torus.

For a subset $J \subseteq \{1, \ldots, k\}$, put

$$n_J = e \prod_{i \notin J} a_i, \qquad m_J = \tau_e \prod_{i \notin J} \ell_i.$$

Then

$$\bigcap_{j \in J} S_j = \mathrm{Sz}(2^{n_J}), \qquad K \cap \bigcap_{j \in J} S_j = C_{m_J} : C_4. \tag{6}$$

The first is the standard coefficient-field intersection. For the second, T_e is always retained. The ℓ_i-factor is retained if $i \notin J$. If $i \in J$, it cannot occur: by (4) it would divide $\gcd(\tau_{ea_i}, \tau_{n_J}) = \tau_e$, contrary to its choice. Binary r belongs to every subfield group, so the same formula holds for all four cosets of the cyclic subgroup.

The RWPRI condition now follows for every proper subset of the $k + 1$ types.

- Before K is selected, select an unused S_i whenever one remains. The current and next groups are simple standard Suzuki subfield groups of relative field degree a_i, so the next is maximal. The lower field exponent remains at least $e \geq 3$.
- When all S_i have been selected, selecting K gives $T_e : C_4$, a full nonsplit torus normalizer and maximal subgroup of $\mathrm{Sz}(2^e)$.
- Once K has been selected, selecting any unused S_i removes exactly one factor ℓ_i, giving a subgroup of prime index.

These uses of subfield and torus-normalizer maximality are retained classical Suzuki facts, not consequences of the disputed source list.

For flag transitivity, write

$$V = \prod_i C_{\ell_i}, \qquad B_0 = T_e : C_4.$$

Then $K = V \rtimes B_0$, and (6) gives exactly the coordinate stabilizers in (3). Lemma D and Theorem C apply. This completes the geometry and flag-transitivity parts of Theorem A.

7 Firmness, residual connectedness, and the base-two exception

The common intersection of all parabolics is $B_0 = T_e : C_4$. A panel missing S_i has size ℓ_i. A panel missing K, with $q = 2^e \geq 8$, has size

$$|\operatorname{Sz}(q) : B_0| = q^2(q-1)\tau'_e/4 \;\geq\; 560.$$

The inequality follows from $\tau'_e \geq q + 1 - \sqrt{2q}$ and its value five at $q = 8$; all factors in the lower bound increase for $q \geq 8$. Each ℓ_i is an odd prime, hence at least three. Thus the flag-transitive geometry is thick.

For residual connectedness, flag transitivity reduces all residues to their standard subgroup systems. The incidence graph of a coset system with at least two types is connected when its marked subgroups generate the ambient residue group. This generation holds uniformly:

- If K was already selected, at least two unused field types remove different prime factors. Their two subgroups together contain every cyclic factor and B_0, hence generate the residue.
- If K is unused and at least two field types remain, their two distinct maximal simple subfields generate the current Suzuki group.
- If only one field type and K remain, the simple base subfield is maximal in the current Suzuki group. The cyclic-by-four subgroup has an extra ℓ_i-factor, so is not contained in that subfield, by (6). Together they generate.

These cases include the ambient incidence graph. They prove residual connectedness in every rank.

The earlier rank-four example has $e = 1$, so the last simple-subfield step cannot be used: $\operatorname{Sz}(2)$ is not simple and is not being declared maximal in $\operatorname{Sz}(2^{a_i})$. Instead its chosen $\tau_{a_i} = 13, 41, 113$ are full prime-order nonsplit tori for $a_i = 3, 5, 7$. At that step select the full torus normalizer, which is maximal. The remaining C_4 has index five in $\operatorname{Sz}(2)$, and the other panel indices are 13, 41, 113. The rank-two residue with one field type and K is connected because its full torus normalizer is maximal and the order-twenty base group contains an element of order five outside it. All other arguments above apply unchanged.

Thus the rank-four example is thick, residually connected, flag-transitive, and RWPRI. This is a proof supplement to the immutable earlier record, whose "unproved" statements correctly describe what had been established at that earlier stage.

8 Retained theory and scope

The substantial external inputs are Lang–Steinberg surjectivity, the standard exceptional Suzuki endomorphism/model, cyclicity of the finite nonsplit tori, and the classical maximality theorems for their normalizers and prime-degree simple subfields. The binary seed, character-lattice calculation, finite descent argument, typed Helly theorem, coordinate attachment, prime-choice argument, and all-subset RWPRI/residue calculations are given here.

The result is an existence and flag-transitivity theorem for explicitly aligned families. It does not say arbitrary subgroups with the same isomorphism types have these intersections. It does not classify all Suzuki geometries, prove a geometric intersection-property axiom, or restore the old candidate completeness claims. Classical ingredients are credited as such, and novelty of this particular assembly remains subject to a fuller literature and external mathematical review.